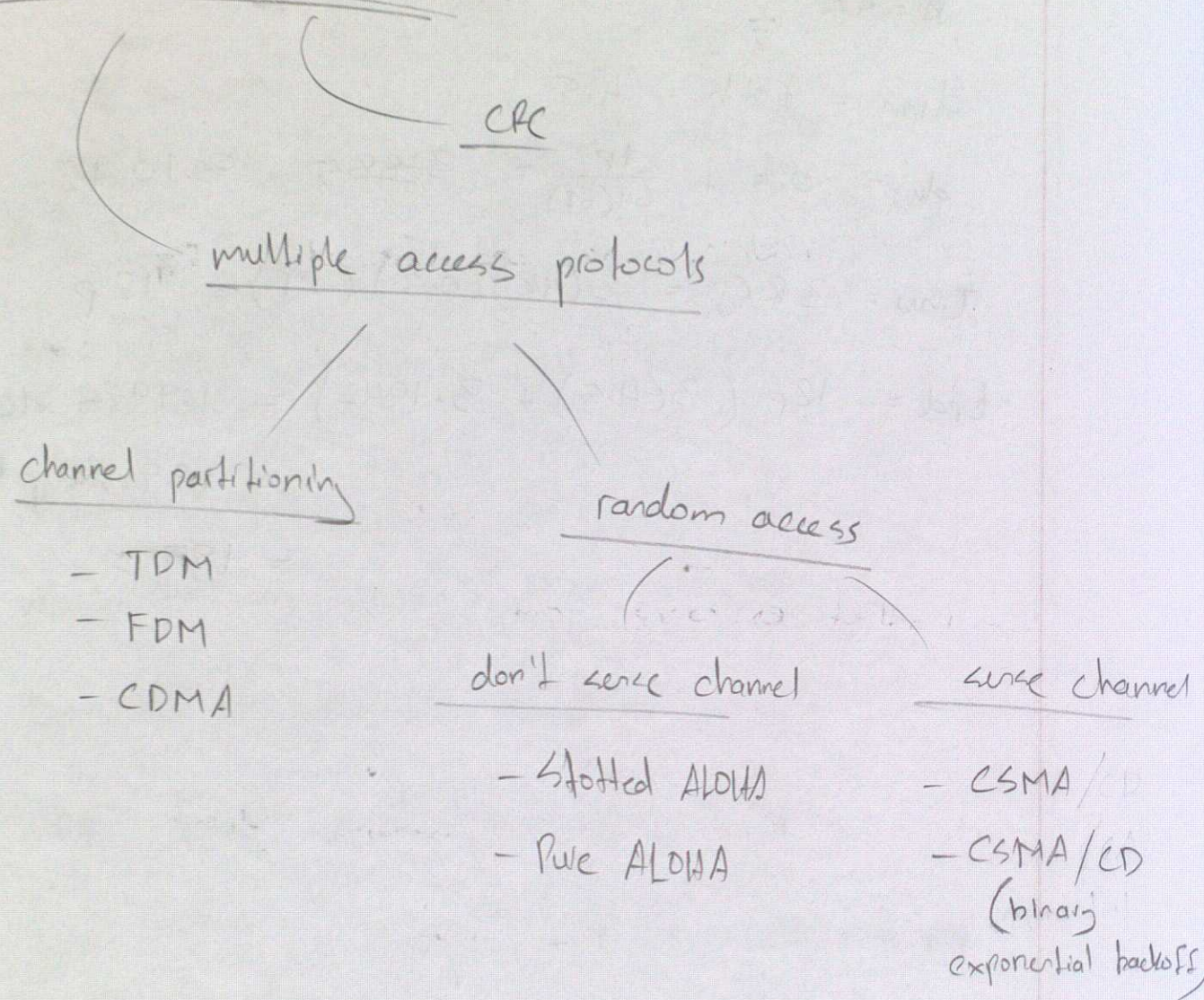


Chapter 6 : Link Layer



Networks Final

Lec 2

$$d_{\text{transmission}} = \frac{L}{R}$$

Lec 3

- packet switching, store and forward $(2 \times \frac{L}{R})$, $2(T + \tau)$
- packet switching, cut through $(1 \times \frac{L}{R})$, $T + 2\tau$
- circuit switching
- packet switching allows more users
- internet exchange points

Lec 4

$$d_{\text{total}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{transmission}} + d_{\text{prop}}$$

$$\text{traffic intensity } \rho = \frac{L \alpha}{R}$$

$$\text{average queuing delay } d_{\text{queue}} = \frac{\rho L}{(1 - \rho) R}$$

- bottleneck link

- layered architecture and encapsulation

Lec 5

- client-server paradigm: HTTP, FTP
- IP address + Port number
- Transport Layer Services: TCP, UDP

Lec 6

- HTTP (TCP, stateless)
 - nonpersistent 1.0 ~ one object per TCP connection
 - persistent 1.1 ~ multiple _____ / _____
 - ↳ pipelining
 - ↳ no pipelining
- parallel TCP connections
- HTTP request: GET, POST, HEAD, PUT
- HTTP response: OK, Moved Permanently, Bad Request, Not Found

Lec 7

- cookies: maintaining user/server state
- web caches/proxy servers: reduce delay (hit ratio)

- conditional GET: only send if modified, else use cached
- email: user agents, mail servers, SMTP

Lec 8

- HTTP/2: divide into frames and schedule (mitigate HOL block)
- HTTP/3: uses UDP, security, error, congestion features
- DNS: application-layer protocol for address resolution
 - root DNS, top-level domain DNS, authoritative
 - host contacts local DNS, which resolves address and returns
 - DNS uses UDP and $\Rightarrow$ caching

(hostname, IP, A, TTL)

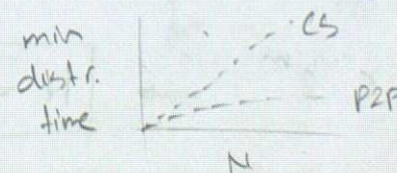
(alias name, canonical name, CNAME, TTL)

(domain, authoritative NS for domain, NS, TTL)

(name, mailserver name, MX, TTL)

Lec 9

- CS scales linearly with N , P2P much slower



review
midterm
Q

- distributed hash table (DHT) ~ (key, value)
(content, IP)
- shortcuts, peer churn

Lec 10

- UDP: if dest port same, everything sent to same socket
- TCP: uses 4-tuple to sent datagrams to respective sockets
(even if same dest port, use other fields to demultiplex)
- UDP: no congestion ctrl → can work through congested network
- HTTP/3 on UDP has reliability and cong. ctrl @ application lyr
- UDP checksum: add everything w/ wraparound, flip, and place

Question
P →

Lec 11 (ARQ: automatic repeat request)

- SW ARQ:
$$\text{utilization}_{\text{sw}} = \frac{L/R}{L/R + RTT}$$

Lec 12

- GBN ARQ:
$$\text{util}_{\text{GBN}} = \frac{N \cdot L/R}{L/R + RTT}$$

$$\left. \begin{array}{l} \text{sender: } \boxed{} \quad N \\ \text{receiver: } \boxed{} \quad 1 \end{array} \right\} N+1 \leq 2^m \rightarrow \boxed{N = 2^m - 1}$$

Lec 13

- SR ARQ: receiver can now buffer out-of-order pkts

$$\left. \begin{array}{l} \text{sender: } \boxed{} \quad H_s \\ \text{receiver: } \boxed{} \quad H_r \end{array} \right\} H_s + H_r \leq 2^m \xrightarrow{\text{if}} N \leq 2^{m-1}$$

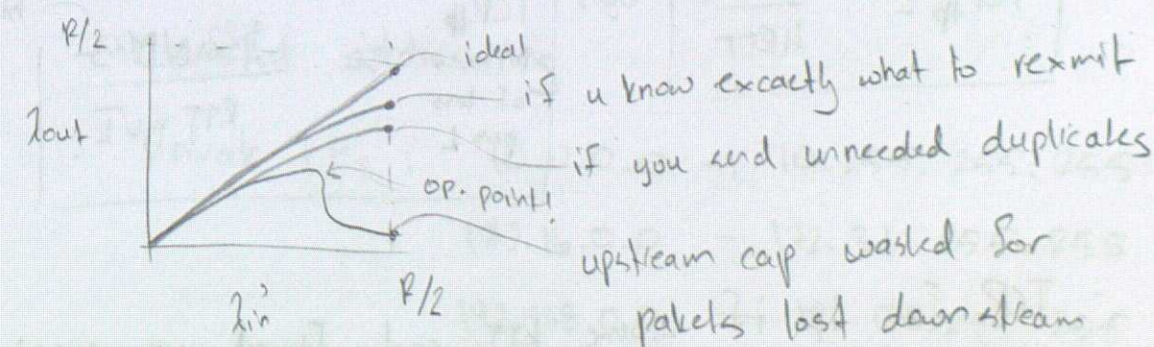
$H_s = H_r = N$

Lec 14 - TCP

- Seq #: bytestream number of first byte in segment
- Ack #: seq # of next byte expected
- Karn's Alg: 1) don't use ack of retransmitted pkt for timeout.
2) insted, double prev. timeout value.
- TCP 3-way handshake

Lec 15

- TCP flow ctrl: receiver advertizes free buffer space in round
- λ_{in} what application layer sends in, λ_{in} what transport layer sends in the network



Lec 16

- AIMD: additive increase, multiplicative decrease
- start at 1 MSS and double every RTT $\leftarrow$ slow start
- once you reach $ssthresh$, increase linearly $\leftarrow$ congestion avoidance

TCP Tahoe

- timeout: slow start
 - $ssthresh = \frac{cwnd}{2}$
 - $cwnd = 1$
- 3 acks: slow start
 - $ssthresh = \frac{cwnd}{2}$
 - $cwnd = 1$

TCP Reno

- timeout: slow start
 - $ssthresh = \frac{cwnd}{2}$
 - $cwnd = 1$
- 3 acks: fast recovery
 - $ssthresh = \frac{cwnd}{2}$
 - $cwnd = ssthresh + 3$

- TCP Cubic: slow down as you approach limit (so can spend more time at best/desired region)

$$\boxed{TCP_{tp} = \frac{3W}{4RTT}} \quad \text{Byts} \quad \boxed{TCP_{tp} \text{ in tens of loss prob } L = \frac{1.22 \times MSS}{RTT \times \sqrt{L}}} \quad \text{bits}$$

- TCP fair tp if same RTT and fixed # sessions in congestion avoidance.

Skipped
quick
slides.

Lec 17

- network layer: forwarding and routing
- router: inputs, switching fabric, outputs.
- dest.-based flooding: longest prefix matching.

Lec 18

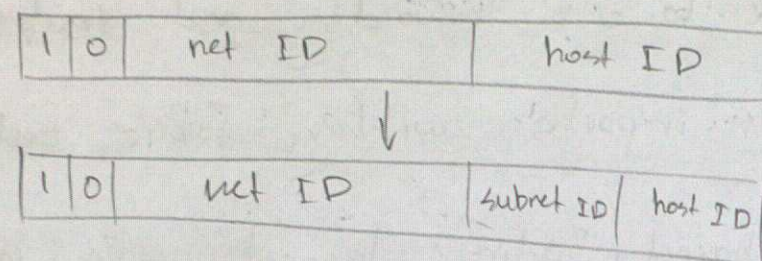
- switching rate $N \times R$ desirable
- switching fabrics: memory, bus, interconnected network
 - memory bandwidth
 - bus contention
- import and output queuing, buffering, scheduling

$$\boxed{WFS = \frac{w_i \times R}{\sum_j w_j}}$$

Lec 19

- Classful addressing
- Private IP's: 10.0.0.0 — 10.255.255.255
172.16.0.0 — 172.31.255.255
192.168.0.0 — 192.168.255.255
- subnet mask to find subnet address

- can partition (eg large class B) network:



Lec 20

- subnet example, distributing large blocks

Lec 21

- short term: CIDR (classless interdomain routing), private IP blocks
- long term: IPv6
- class A, B for clear need, instead chop up and allocate necessary contiguous blocks of class C addresses
- supernetting: summarize contig. blocks of class C addresses under variable length mask.
- CIDR: a.b.c.d/x ← * bits in subnet part
 - deals with router explosion problem
 - reflects physical topology of network
- DHCP: discover, offer, request, ack
UDP

Lec 22

- CIDR address aggregation
- NAT: network address translation
- address mapping
 - static: $\{ \text{---} \} \leftrightarrow \{ \text{---} \}$
 - dynamic: $\{ \text{---} \} \leftrightarrow \{ \text{---} \}$
- overloading (map all to one IP but diff port nums)
- NAT traversal problem solved by
 - static NAT configs
 - UPnP IGD protocol

Lec 23

- IPv6: no checksum, fragmentation, or options (∴ fixed 40B header)
- tunneling: IPv6 datagram carried as payload in IPv4 datagram
- forwarding
 - dest. based forwarding
 - generalized forwarding: SDN, OpenFlow (match + action)

Lec 24

- network control plane
 - per-router-ctrl (trad)
 - logically centered ctrl (SDN)
- link state: global
- distance vector: decentralized

- link state:

- quick recovery from link failure
- watch out for old update mssgs
- oscillations possible (when cost is $f(\text{traffic})$)
- different costs in 2 directions

Lec 25

- Distance vector: Bellman Ford equation

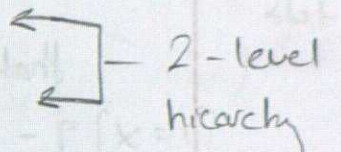
Lec 26

- DV issue poisoned reverse (if a node is the next node in path to dest, advertise ∞ to it)
(basically, advertise ∞ downstream to dest)
- DV takes time to converge, can have loops, count to infinity, and can have black-holing (Pakistan incident)
- LS preferred bc fast, loopless, supports multiple paths
- ICMP over IP: network layer protocol for error report, ping.
 - used by traceroute.

use src IP
in IP header

Lec 27: border/gateway routers do both inter and intra-AS routing

- AS: autonomous system controls group of routers
 - stub, multihomed, transient
- OSPF: intra-AS link state routing protocol (uses IP directly)
 - local area routers flood local area
 - area border routers flood backbone



Lec 28

- BGP: propagate reachability information (sits on top of TCP)
 - eBGP to obtain reachability from outside AS
 - iBGP to propagate reachability inside AS
- inter-AS: policy (BGP big on policy)
- intra-AS: performance (would pick lowest intra-routing cost instead of worrying about inter-AS costs)

Lec 29

- CRC bits 1 bit less than generator

- 1) shift D by $(G-1)$ bits to make room for CRC
- 2) divide D by G and put remainder in CRC and send
- 3) receiver divides result by G
 - if 0, then no error
 - else, error, discard quietly

P check
register
clk

Lec 30

- to be catchable, error should be indivisible by g
- for 1-bit error, ensure $g(x)$ has $+1$
- for 2-bit error n bits apart, must have that $e(x) = x^n + 1$ is indivisible by $g(x)$
- $g(x)$ is a primitive poly order m iff the smallest n s.t. $x^n + 1$ is divisible by $g(x)$ is $n = 2^m - 1$

eg $g(x) = x^2 + x + 1$ is primitive, $m = 2$
 $n = 2^m - 1 = 2^2 - 1 = 4 - 1 = 3$

→ $x^3 + 1$ is divisible by $g(x)$ → ∴ cannot catch
 ∴ errors 2 bits apart or closer are detectable

⇒ ∴ select pkt length as $n < 2^m - 1$ if you have $g(x)$ primitive order m in order to catch all 2-bit errors

- odd * errors → $g(x)$ should have $x+1$

Lec 31

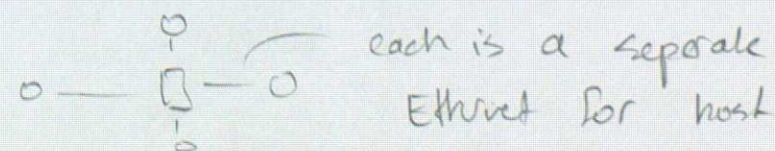
- slotted ALOHA: efficiency $\frac{1}{e}$ binomial
- $P[x \text{ packets in one slot}] = P[x=k] = \binom{N}{k} p^k (1-p)^{N-k}$
- $P[x=0] = (1-p)^N$ idle slot
- $P[x=1] = N p (1-p)^{N-1}$ success slot
- $P[x \geq 2] = 1 - P[x < 2] = 1 - P[x=0] - P[x=1]$ collision slot
- $P[\text{a given node has success}] = P_s = p(1-p)^{N-1}$
- time to success geometric, ∴ equal to $\frac{1}{P_s}$

Lec 32

- CSMA: 1-persistent, p-persistent, non-persistent
- Ethernet has binary exponential backoff
 (pick from $[0, \dots, 2^m - 1]$ times 512 bit times after m th collision)

- CSMA/CD efficiency = $\frac{1}{1 + 5 \frac{t_{prop}}{t_{trans}}}$

- switch reduces collision domain



- Ethernet has type (upper layer protocol)
 - 802.3 has length
 - if $\square \leq 1500 \rightarrow \text{length} \therefore 802.3$
 - if $\square \geq 1536 \rightarrow \text{type} \therefore \text{Eth}$
 - length of frame detected in Eth by silence
 - Eth MAC protocol: unslotted CSMA/CD, binary backoff
+ if collision detect, abort + jam
- 48 bits

Lec 33

- 48-bit MAC address: unicast, broadcast, multicast
- ARP: address resolution protocol to learn MAC addrs
- use layer 2 (MAC) on each link (ARP tables)
- man in the middle attack

★ review broadcast MAC address

Lec 34

- hub (repeater) ~ Phys. layer
- switch (bridge) ~ Link layer, selective fwd/flood, transparent, plug n play
- switch self-learns and fills switch table
- ★ write comp-contrast routers and switches

Lec 35

- 1 - port-based VLAN
 - trunk port: connect VLANs defined over multiple switches
- 2 - tagged VLANs
 - multiprotocol label switching (MPLS)

P
review